Rebotic — Hardware Parts List (rev 2)

Conversational expo robot · laptop = mind · body moves: neck, head, two arms (2-axis each), LED eyes · wired USB audio · software push-to-talk (no hardware button)

Prepared 2026-07-24 · matches body/specs/03-electronics.md (pending update) · prices are approximate USD — local market prices vary

> **Changed from rev 1:** removed the jaw servo + jaw pushrod linkage (no mouth movement) and the arcade push-to-talk button (redundant — push-to-talk is already handled in software, see `hearing.py`). Added a second servo per arm so each arm has two independent axes ("moving all around" instead of just up/down).

1 · Controller (the body's motion brain)

Part (market name)	Qty	Description	Purpose in the robot	Approx.
ESP32 DevKit V1 (30-pin) *search: "ESP32 DevKit V1 WROOM-32 development board"*	1	Small WiFi-enabled microcontroller board with USB port. Any ESP32-WROOM-32 dev board works.	Sits inside the body. Receives mood commands from the laptop over WiFi and drives all servos and eyes.	$5–8

2 · Movement (neck, head, arms)

Part (market name)	Qty	Description	Purpose in the robot	Approx.
SG90 9g micro servo *search: "Tower Pro SG90 9g micro servo motor"*	6 + 2 spare	Tiny hobby servo, ~180° rotation, plastic gears. Comes with horns and mounting screws.	One each for: neck pan (turns left/right), head tilt (nods), left arm lift, right arm lift, **left arm rotate (new)**, **right arm rotate (new)** — the rotate servos let each arm swing forward/back instead of only up/down (wave, raise, open gestures). Spares are mandatory — plastic gears strip on long demo days.	$2–3 each
Pan/tilt servo bracket kit (SG90 size) — neck *search: "SG90 pan tilt bracket kit 2 axis"*	1	Two-axis plastic mounting frame that holds two SG90s stacked.	The neck mechanism — bottom servo turns the head, top servo nods it.	$3–5
Small servo L-bracket (SG90 size) — shoulders (new) *search: "SG90 servo mounting L bracket"*	2	Small right-angle bracket that mounts one servo's horn onto a second servo's body, at 90°.	One per shoulder — mounts the new rotate servo perpendicular to the lift servo, so each arm gets two independent axes off a single shoulder point.	$1–2 each

Removed from rev 1: *Pushrod / stiff steel wire* (jaw linkage) — no longer needed, there's no jaw.

3 · Eyes & input

Part (market name)	Qty	Description	Purpose in the robot	Approx.
WS2812B 8-LED ring (5V) *search: "WS2812B 8 bit LED ring 5V NeoPixel"*	2	Ring of 8 individually-addressable RGB LEDs, single data wire, chainable.	The eyes — blink randomly, dim cyan at rest, bright white when attending, pulsing amber while thinking, eye-roll on shrug.	$2–4 each

> Removed from rev 1: *Arcade push button, 60mm* — push-to-talk stays software-only (keyboard PTT already in `hearing.py`); no physical button.

4 · Ears & voice (USB audio — plugs into the LAPTOP, never the ESP32)

Part (market name)	Qty	Description	Purpose in the robot	Approx.
USB directional microphone *search: "USB cardioid gooseneck microphone" or "mini USB conference mic cardioid"*	1	Compact plug-and-play USB mic. Cardioid/directional pattern is the priority — it hears what's in front, rejects room noise. Worth spending extra here.	The robot's ears — mounted in the head/chest facing the visitor. The laptop listens through it via a USB cable.	$10–25
USB-powered mini speaker, 3–5W *search: "USB powered mini speaker 5V wired" (USB sound, or 3.5mm audio + USB power)*	1	Small full-range wired speaker powered over USB. 3–5W is plenty for one visitor at arm's length.	The robot's voice — mounted behind a grille in the chest.	$8–15
USB 2.0 extension cable, 3m (A-male to A-female) *search: "USB 2.0 extension cable 3m"*	2	Plain USB extension leads.	The hidden "tubes" — route mic and speaker from the body down the stand to the laptop. Plug into two separate laptop ports (not one hub).	$3–5 each

5 · Power

Part (market name)	Qty	Description	Purpose in the robot	Approx.
5V 5A DC power adapter (mains) *search: "5V 5A power supply adapter 5.5mm barrel jack"*	1	Wall adapter, 5 volts at 5 amps, barrel plug.	Dedicated servo + LED power rail. **Six** stalled servos now pull ~4A (was ~3.3A with 5) — 5A is still the floor but the margin is tighter than rev 1. Never power servos from the ESP32 board.	$8–12
DC barrel jack to screw-terminal adapter (female) *search: "DC 5.5x2.1 female barrel jack screw terminal adapter"*	1	Lets you connect bare wires to the adapter's barrel plug, no soldering.	Taps the 5V supply into the servo power distribution wiring.	$1–2
Electrolytic capacitor, 1000µF / 10V *search: "1000uF 10V electrolytic capacitor"*	1 (buy 2–3)	Large can-shaped capacitor. Mind the polarity stripe when wiring.	Sits across the servo power rail — absorbs the current spike when servos start moving so nothing resets.	<$1
Resistor, 470Ω, 1/4W *search: "470 ohm resistor 1/4W"*	1 pack	Ordinary small resistor.	Goes in series with the eyes' LED data line — protects the first pixel from signal spikes.	<$1
Micro-USB data cable *search: "micro USB data cable" (must be DATA cable, not charge-only)*	1	Standard phone-style cable — verify it carries data.	Powers the ESP32's logic from the laptop and is how firmware gets flashed onto it.	$2–3

6 · Assembly & body shell

Part (market name)	Qty	Description	Purpose in the robot	Approx.
Dupont jumper wire set (M-M, M-F, F-F) *search: "dupont jumper wires 120pcs kit"*	1 kit	Pre-crimped hookup wires for breadboard/header connections.	All wiring between ESP32, servos, and LEDs. Buy the big kit — spares are on the expo ship list.	$4–6
Breadboard, half-size (400 tie points) *search: "solderless breadboard 400 points"*	1	Solderless prototyping board.	Holds the ESP32 and the power distribution — no soldering needed for the whole build.	$3–5
Foam board / foam-core sheets, 5mm *search: "foam board A3 5mm" (art/stationery shop)*	2–3 sheets	Lightweight rigid board, cuts with a knife.	The head shell and both arms. **Must stay light** — each arm now carries two servos instead of one, so keep the arm panels as thin/small as the added weight allows.	$3–6
Acoustic / packing foam scrap *search: "acoustic foam panel small" (or reuse packaging foam)*	small piece	Soft open-cell foam.	Isolation pad between speaker and mic mounts — stops the robot hearing its own voice and interrupting itself.	$0–3
Hot glue gun + sticks, zip ties, electrical tape *(household / any hardware shop)*	—	General assembly supplies.	Mounting servos in foam, strain-relieving the cables where they exit the body.	$5–8

> **Estimated total: ~$58–95** · The USB microphone is still the one place to spend more — a genuinely directional mic is the biggest single upgrade to demo quality in a noisy hall.

Buying tips

- **Already owned, not on the list:** the laptop (the mind), a phone (WiFi hotspot), and a wired headset mic (expo fallback input).
- **Servos:** buy all 8 (6 + 2 spare) from the same seller/batch so they behave identically. "MG90S" (metal gear) is a worthy ~$1/pc upgrade for the four arm servos if available — they now carry more duty cycle with two axes moving per arm.
- **Speaker:** if it has a 3.5mm audio plug + USB power plug instead of pure USB audio, that's fine — the laptop has both sockets.
- **Power adapter:** don't buy 5V 3A to save two dollars — 5A is the floor, and it's tighter now with 6 servos instead of 5. If budget allows, a 5V 6A adapter buys back the headroom rev 1 had.
- **Capacitor polarity:** electrolytic caps have a marked negative stripe — wiring one backwards destroys it.
- **Micro-USB cable:** test it transfers data (connect a phone) — charge-only cables are the classic "board won't flash" trap.

Rebotic body subproject · full wiring, pinout and power details: body/specs/03-electronics.md (still describes the rev-1 jaw/button build — needs updating to match this list) · assembly order: body/specs/04-bringup.md